Melusi Population Estimation using U-Net and Random Forest Regression

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ABSTRACT

This research aims to develop a methodological approach for estimating small-area population in Melusi, an informal settlement in Atteridgeville, Pretoria West, using high-resolution drone imagery and machine learning. This paper presents the exploratory data analysis (EDA) undertaken to assess the scope, structure, and limitations of the available datasets prior to model development. The primary dataset which is a 3 cm resolution orthophoto; was inspected through metadata extraction, windowed pixel sampling, per-band histograms, inter-band correlation analysis, and a brightness thresholding experiment. Spectral analysis of the RGB bands revealed sufficient colour variation to distinguish surface types, though a baseline thresholding test demonstrated that naive spectral methods overestimate rooftop coverage by approximately double due to spectral overlap between soil and metallic roofing, reinforcing the need for the proposed U-Net deep learning architecture. The LiDAR point cloud was assessed through header inspection and elevation profiling, revealing the terrain gradient across the settlement with an unclassified, intensity-absent point cloud. The GRT-INSPIRED household survey was filtered to the Melusi subset (8174 individuals) and its structure examined for linkage potential to the imagery. A critical limitation was identified: the survey lacks dwelling-level coordinates or identifiers, preventing direct spatial linkage between demographic records and the orthophoto. All analysis was conducted in Python using a Google Colab environment with geospatial libraries including rasterio, laspy, and geopandas. The full reproducible analysis is available in the accompanying Jupyter Notebook.

Keywords

Exploratory Data Analysis, Image Processing, Population, Informal.

1. INTRODUCTION

Melusi settlement in Atteridgeville is a relatively new informal settlement in the City of Tshwane with a population density of approximately 10300 people per square km in 2023 [1]. Currently, most popular methods of accumulating census data in informal settlements in South Africa is through community driven self-enumeration through various organisations [2]. This proves to be a challenge as the settlements are frequently left out of city planning due to lack of detailed information [2].

The purpose of this research is to investigate whether it is possible to develop a reproducible methodology that will accurately predict informal settlement population using only dwelling size data. The research model will be built on data provided for Melusi settlement. The model will be two-stage ensemble prediction starting with a U-Net deep learning algorithm to identify individual dwellings, thereafter, a Random Forest regression model will be trained using the dwelling outline data as well as the census data to estimate how many people live in an area based on the number, size, and density of the detected dwellings.

This report serves to provide the details of the exploratory data analysis (EDA) of the various data sources made available. The EDA will determine if there is sufficient data available to support the desired model pipeline described above. Any issues in data quality, continuity, and transposability will also be outlined in detail in this report as well as in the accompanying Jupyter Notebook which is linked in chapter 7. It is also important to note that majority of the results of the EDA process are presented in the Notebook, and this report presents high level findings, concerns, and methods to proceed.

The structure of this report follows the EDA checklist outlined by Peng and Matsui (2018), adapted for geospatial raster data. Section 2 describes the available datasets and their properties. Section 3 provides a high level data description of the sources. Section 4 presents the detailed analysis, covering image quality assessment, spectral distributions, a baseline thresholding experiment, and LiDAR terrain profiling. Section 5 discusses the findings, identifies limitations, and outlines the next steps required to proceed to the modelling phase. The accompanying Jupyter Notebook (01_eda.ipynb), committed to the project GitHub repository, contains the full reproducible analysis with all code, outputs, and figures and is available in Section 7.

2. INITIAL EXPLORATION

In order to fully grasp the scope of the data provided, a system approach had to be adopted and agreed upon before any EDA commenced. This was achieved by breaking up the analysis into three main sections. First was exploring the descriptions and metadata of the datasets by answering the question below:

1. What does the imagery cover?
2. What is the spectral profile of the dwelling's vs non-dwelling's
3. What are the data types and what requirements should be in place for the environment such that visual depictions of the data can be obtained.

The second step was to explore the datasets based on the results of the metadata obtained above. This was done to identify any spatial patterns in dwelling densities across the settlements. Additionally, this assisted in identifying unique linking features that will be used to merge the different data sources.

Finally, an investigation was conducted into which of the various features can be used to accurately provide an estimate of the population of this area. This was achieved by using image-derived features as well as features from the survey data which provided the true population size of the area.

3. DATA DESCRIPTION

Five datasets were assessed. The primary data set is a single areal image captured by drone of the entire Melusi settlement. Anciliary

datasets include a compressed version of the image for quick reference, a 3D point cloud generated by LiDAR. The set is completed, by household survey data collected by a separate study. The datasets are highlighted in Table 1 below.

Table 1 Dataset Summary

Dataset	Format	Size	Key Properties
Orthophoto	GeoTIFF	7.03 GB	102730 x 41815 pixels sampled at 3cm resolution in 4 bands (RGB and Alpha) in Lo29
	ECW	675 MB	
Ground Control Points	Text	4 Points	Ground positioning survey points used to verify orientation.
LiDAR Point Cloud	LAS	5.9 GB	296 million points, 3D laser scan of the terrain and structures. Lo29 unclassified
Household Survey	Excel	49 MB	8174 points for Melusi. GRT-Inspired baseline demographic study of dwelling type, household size, spatial coordinates

3.1 Spatial Validation

The four GCP coordinates points are surveyed positions on the ground whose precise locations are known. These were used to confirm that the image is correctly georeferenced by:

a. Confirming the GCP's are contained in the bounds of the GeoTIFF file.
b. Visually plotting the GCP's to understand the spread across the map.

Failure to validate this stage would have meant that feature identification of the GeoTIFF was either not processed correctly, or the there was a critical coordinate mismatch, meaning that the image could not reliably be linked to any real-world locations.

3.2 Data gap

The initial household data that was published publicly and made available to the team had survey records of 8174 individuals across Melusi with detailed demographic and dwelling data. However, it was later noted that the dataset did not contain any coordinate reference, street address, or unique identification for an individual to a respective dwelling. This posed a major limitation in the expected data structure and model architecture, which required a direct link between the GeoTIFF and the survey data. However, a coordinate file was requested from the project partner, and this file has since been made available. The contents of this file are strictly private, however, the data analysis is still covered in chapter 4 below.

4. DETAILED ANALYSIS

This chapter will highlight the key findings that were obtained when EDA was conducted on each dataset. It critically important to note that this chapter should be read alongside the EDA Jupyter Notebook where further detailed statistical and visual findings are presented.

4.1 GRT-I Baseline data

When we explored the GRT-I Baseline data [3], we found that there is no unique identifier for the response to link the feature to a unique dwelling. After filtering for Melusi we get 8174 responses with 278 columns (features). Deep diving into the data, we found that the total resident population is 21238 and the combined population that includes the non-residence is 21486. Based on the description of the survey and after checking if there are duplicates responses 0 rows have the same meaning all 8174 responses are unique. As a result, assuming that all responses are from different people from different dwellings and the survey covers the whole or most of Melusi, then the number of dwellings in Melusi is 8174. From the histogram in the notebook we can see that most dwellings have less than 4 people living in each of the dwellings.

4.2 Image Quality and Coverage

The GeoTIFF orthophoto was examined at multiple scales. A compressed ecw file overview of the entire image revealed the full extent of the Melusi settlement, including dense clusters of dwellings, stretches of non-dwelling land (vegetation, parks, forests), open ground, and what appear to be water bodies. The image covers approximately 3.1x1.25 km. A zoomed and crop (60x60m) confirms that at 3cm resolution, individual dwellings and other fine details such as pathways and vehicles are clearly visible. This level of detail is sufficient for a deep learning model to learn the visual characteristics of dwelling rooftops.

The image metadata analysis found no "nodata" value, however, empty areas at the image boundaries were found to be filled with pure white pixels (value 255 in all three colour channels). The fourth (alpha) band, which was expected to indicate valid versus empty regions, contains only zeros. This means that all downstream processing must treat pixels where Red = Green = Blue = 255 as empty, or risk inflating dwelling counts with false detections in blank areas.

4.3 Spectral Analysis

Each pixel in the image is described by three colour values: Red, Green, and Blue, each ranging from 0 (dark) to 255 (bright). These distributions were plotted in order to understand what surfaces dominate the scene and whether different land cover types are distinguishable by colour alone.

The Red band shows a bimodal distribution. The green band peaks higher than red, and blue is notably lower. The correlation between the red and blue bands is 0.898, meaning that the bands are both well related and interface at different pixels.

4.4 A Baseline Test

Before conducting complex U-Net deep learning models, a simple test was done to investigate the possible approach:

Applying different brightness and colour thresholds to separate rooftops from background.

This establishes a baseline against which the deep learning model's performance can later be measured.

The results showed that brightness-based thresholding flagged 34% of pixels as "rooftop". This was roughly double the expected coverage of 15 - 25%. This seems to occur due to sandy soil and metallic rooftops having similar brightness bands. A threshold fails to tell them apart. However, a vegetation index cleanly identified 23% of the area as green cover, because vegetation has a genuinely distinct colour signature (high green, low red and blue).

This finding directly justifies the use of a U-Net deep learning model for dwelling extraction. Unlike a simple threshold that examines each pixel independently, the U-Net analyses spatial patterns which enables it to distinguish a dwelling from a similarly bright patch of dirt.

4.5 LiDAR Analysis

The LiDAR dataset is a 3D scan of the settlement produced by the same drone flight. Each of the 296 million points records an X, Y, and Z coordinate, where Z is the elevation above sea level. The elevation heatmap reveals a terrain that slopes approximately from the north to the south. This topographic variation means that building height cannot be inferred from raw elevation alone.

A suggested approach would be to use a normalised height model which would subtract the ground surface from Z values. This will be needed to isolate structure heights.

The point cloud is currently unclassified, and the laser intensity channel contains only zeros. Classification into ground, building, and vegetation categories will be required before height-based features can be extracted for the Random Forest model.

5. DISCUSSION

This exploratory analysis confirms that the Melusi dataset can infact be used for machine learning based population estimation, with a few considerations that should either be addressed or assumed before modelling commences.

5.1 Orthophoto

The orthophoto provides exceptional detail. At 3cm resolution, dwellings are unambiguously visible and spectrally distinguishable from surrounding surfaces. The colour variation across the three bands provides sufficient spectral information for a deep learning model to learn unique discriminative features.

5.2 Discriminative Methods

Simple methods are insufficient. Brightness thresholding overestimates roof coverage by roughly two times because in the sample image selected, the soil and metal roofs seem to overlap spectrally when brightness is isolated. This gap further motivates the U-Net architecture, which leverages spatial context rather than pixel values alone, but reverse comparison of convoluted resultant image with original image.

5.3 Survey Linking

The critical bottleneck is the spatial link between survey and imagery. Without detailed dwelling unique identifiers, population modelling is limited to zone-level aggregate estimates rather than per dwelling predictions. Securing this link is the highest priority next step.

5.4 LiDAR Support

LiDAR offers supplementary value. Building heights derived from the point cloud can serve as additional features in the Random Forest, but the data requires classification and normalisation before use. Immediate next steps include:

(1) Obtaining a reliable dwelling link file for the survey data to imagery
(2) Tiling the orthophoto into an optimal resolution for preprocessing training pipeline
(3) Creating labelled training data through manual annotation or transfer from open building footprint datasets
(4) Classifying the LiDAR point cloud to extract building height features.

6. CONCLUSION

The EDA confirms that the available data is sufficient to support the proposed two stage population estimation pipeline. The orthophoto provides high quality dwelling level detail at 3cm resolution, with sufficient spectral variation across the RGB bands to distinguish surface types.

The baseline thresholding experiment validated the choice of U-Net over simpler methods, as brightness alone overestimates roof coverage by approximately double due to spectral overlap between soil and metallic roofing. The LiDAR point cloud offers complementary elevation data but requires classification and ground normalisation before building heights can serve as features in the Random Forest model.

The most critical limitation is the absence of a spatial link between the household survey and the orthophoto. Without dwelling-level coordinates, population modelling is restricted to zone-level aggregate estimates rather than per-dwelling predictions. Securing this spatial link from the project partner is the highest priority next step, followed by tiling the orthophoto for U-Net training, generating labelled dwelling data, and classifying the LiDAR point cloud.

7. GITHUB REPO LINK

The GitHub Repository link for this project is provided below. The Jupyter Notebook were the normative references made in this report are also found in the repo and are linked below:

Table 2 Important Links

What	Link
GitHub Repo	Link
EDA Notebook	Link

8. REFERENCES

[1] N. P. S. Sempe, "A proposed risk assessment framework for public health responses: A case study of Melusi, an informal settlement in Gauteng, South Africa," University of Pretoria, Pretoria, 2023.

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